

METIMMOX-1 Expected Values of Partially Perfect Information (EVPPI) Analysis

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Introduction

This report presents the Expected Values of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 economic model. EVPPI quantifies the maximum value of obtaining perfect information about specific model parameters or groups of parameters, helping to prioritize future research efforts.

The EVPPI analysis identifies which parameters, if known with certainty, would provide the greatest reduction in decision uncertainty. This information is crucial for:

- Prioritizing future research investments
- Identifying parameters requiring more precise estimation
- Guiding the design of additional studies
- Supporting evidence development strategies

2 Methods

2.1 EVPPI Calculation Methodology

The EVPPI for parameter θ is calculated as:

$$EVPPI_{\theta} = E_{\theta}[max_j E_{\theta^{-\theta}|\theta}[NMB(j, \theta, \theta^{-\theta})]] - max_j E[NMB(j, \theta)]$$

Where:

- θ represents the parameter(s) of interest
- $\theta^{-\theta}$ represents all other parameters
- j represents treatment strategies
- NMB represents Net Monetary Benefit

2.2 Implementation

The EVPPI analysis uses a nested sampling approach:

1. **Outer loop:** Integrates over the distribution of the parameter(s) of interest
2. **Inner loop:** For each parameter value, estimates expected NMB using simulations with similar parameter values
3. **Calculation:** Computes the difference between expected value with partial perfect information and current expected value

3 Results

3.1 Overall EVPPI Summary

No EVPPI results available for display.

Key statistics not available - missing required objects or empty results.

3.2 EVPPI Visualization

No EVPPI results available for plotting.

3.3 Research Prioritization

No EVPPI results available for research prioritization.

3.4 Parameter Group Analysis

No EVPPI results available for parameter group analysis.

4 Discussion

4.1 Interpretation of Results

Unable to provide interpretation - missing required results or EVPI data.

4.2 Limitations

The EVPPI analysis has several important limitations:

1. **Computational Approximation:** The nested sampling approach provides an approximation of the true EVPPI, which may underestimate values for highly nonlinear models.
2. **Parameter Independence:** The analysis assumes parameters can be learned independently, which may not reflect real-world research scenarios.
3. **Linear Utility:** The analysis assumes linear utility in money, which may not hold for all decision-makers.
4. **Model Structure:** EVPPI values are conditional on the current model structure and may change with model modifications.

5 Conclusions

****No conclusive EVPPI results available.****

This may indicate:

- Very low decision uncertainty in the model
- Insufficient parameter variation in the PSA
- Technical issues with the EVPPI calculation

Report compiled on 2025-10-15 by Ben Geisler using the METIMMOX-1 health economic model.